

Congenital Factor X Deficiency — Comprehensive Disease Characterization Report

Disease: Congenital (Hereditary) Factor X Deficiency — "Stuart–Prower factor deficiency"
MONDO ID: MONDO:0009212 · **OMIM:** 227600 · **Gene OMIM:** 613872 (*F10*) · **Category:** Mendelian (autosomal recessive) **Report basis:** 13 confirmed findings · 46 papers reviewed · 5 investigation iterations. Literature-derived from aggregated disease-level resources plus case reports, cohorts/registries, model organisms, and structural studies.

Summary

Congenital Factor X (FX) deficiency is an ultra-rare autosomal recessive bleeding disorder caused by biallelic loss-of-function or dysfunction-inducing variants in the **F10** gene on chromosome **13q34**. F10 encodes coagulation factor X (Stuart–Prower factor), a vitamin K-dependent serine protease synthesized in the liver that sits at the **convergence point of the extrinsic (tissue factor/FVIIa) and intrinsic (FIXa/FVIIIa) coagulation pathways**. Its activated form, FXa, assembles with cofactor FVa, calcium, and phospholipid membranes into the **prothrombinase complex** that converts prothrombin to thrombin — the enzyme that ultimately forms the platelet plug and fibrin clot. When both F10 alleles are defective, FXa output falls, thrombin generation is impaired, and a bleeding diathesis results.

The clinical severity of the disorder correlates strongly with residual FX coagulant activity (FX:C). Severe deficiency (FX:C <1%) manifests in the **neonatal period** with umbilical-stump bleeding and, most gravely, **intracranial and subgaleal hemorrhage**, whereas milder forms cause mucocutaneous bleeding, epistaxis, easy bruising, menorrhagia, and hemarthrosis. Diagnosis rests on the characteristic pattern of **simultaneously prolonged prothrombin time (PT) and activated partial thromboplastin time (APTT)** — a signature of a common-pathway defect — confirmed by a specific FX activity assay, with FX antigen (FX:Ag) measurement distinguishing quantitative (type I) from dysfunctional (type II) disease.

Management is tiered around **factor replacement**: high-purity plasma-derived FX concentrate (pdFX / Coagadex) is the first single-factor-specific product approved in the US and EU and is safe and effective for on-demand treatment, prophylaxis, and perioperative cover; prothrombin complex concentrate (PCC) and fresh frozen plasma (FFP) serve as alternatives where FX concentrate is unavailable, with antifibrinolytics and hormonal therapy as adjuncts. Prophylaxis prevents the otherwise poor outcome of recurrent early-life intracranial hemorrhage. Prevention centers on genetic counseling, carrier/cascade testing, and prenatal/preimplantation diagnosis, particularly relevant given the higher prevalence in consanguineous populations.

1. Disease Information

Congenital Factor X deficiency is "**a rare autosomal recessive bleeding disorder caused by mutations in the F10 gene located on chromosome 13q34-ter**" (PMID: 30507709). Factor X, originally named **Stuart–Prower factor**, is a plasma glycoprotein that plays a pivotal role in the coagulation cascade.

Key identifiers:

Resource	Identifier
MONDO	MONDO:0009212
OMIM (phenotype)	227600 (Factor X deficiency)
OMIM (gene)	613872 (F10)
Orphanet	ORPHA:328 (Hereditary factor X deficiency)
ICD-10	D68.2 (Hereditary deficiency of other clotting factors)
ICD-11	3B12.0
MeSH	D005166 (Factor X Deficiency)
HGNC	HGNC:3528 (F10)
UniProt	P00742 (FA10_HUMAN)

Synonyms / alternative names: Hereditary factor X deficiency (HFXD), Stuart–Prower factor deficiency, Stuart factor deficiency, congenital FX deficiency.

Data provenance: The evidence in this report is derived from **aggregated disease-level resources** — case reports, multicentre genotype–phenotype cohorts, disease registries (EN-RBD, PRO-RBDD), and clinical trials — rather than from individual-patient EHR extracts.

2. Etiology

Disease causal factors. The disorder is **monogenic and genetic**. It is caused by biallelic pathogenic variants in **F10**, inherited in an autosomal recessive fashion. There is no environmental or infectious cause of the *congenital* form. (A distinct **acquired** FX deficiency exists — associated with AL amyloidosis, autoimmune disease such as Sjögren's syndrome, and transient inhibitors — but this is etiologically separate and is discussed only for differential diagnosis.)

Genetic risk factors. The causal variants are within F10 itself. A multicentre cohort **"identified 22 separate mutations, including 15 missense mutations, 2 deletions, 4 splice site mutations, and 1 nonsense mutation"** among 24 individuals, with 149 F10 mutations reported to date at the time of that study ([PMID: 30507709](#)). Missense variants predominate. Because the disorder is recessive, **carrier (heterozygous) status** in both parents is the fundamental genetic risk factor for an affected child.

Environmental risk factors. **Consanguinity** is the dominant modifiable risk determinant at the population level — the disorder is markedly more prevalent in populations with high rates of consanguineous marriage (Iran, Turkey, Pakistan, Egypt). Consanguineous unions increase the probability that both parents carry the same rare F10 allele; documented pedigrees explicitly attribute homozygous disease to consanguineous marriage (e.g., the p.Val298Met Chinese pedigree, [PMID: 27264807](#)). No toxic, occupational, or lifestyle exposure causes the congenital disease.

Protective factors. No genetic protective alleles or environmental protective factors have been established for congenital FX deficiency. Heterozygous carriers with FX:C 40–50% are typically asymptomatic, effectively conferring a "protected" phenotype relative to homozygotes, but this reflects gene dosage rather than a distinct protective mechanism.

Gene–environment interactions. The principal interaction is between the recessive F10 genotype and the sociocultural environment of consanguinity, which raises homozygosity rates. Vitamin K status and hepatic function modulate overall FX levels (FX is vitamin K–dependent), so intercurrent liver disease, vitamin K deficiency, or vitamin K antagonist exposure can compound a congenital deficiency — an environmental modifier of the biochemical phenotype rather than a cause.

3. Phenotypes

The phenotype is a **hemorrhagic diathesis** whose severity tracks residual FX:C. Bleeding spans mucocutaneous, deep-tissue, and life-threatening central-nervous-system bleeds.

Phenotype	Type	HPO term	Frequency / severity	Onset
Abnormal / prolonged bleeding	Clinical sign	HP:0001892	Universal in severe disease	Neonatal–variable
Epistaxis	Symptom	HP:0000421	11/12 (91%) in a severe cohort	Childhood
Easy bruising / ecchymoses	Sign	HP:0000978	11/12 (91%)	Childhood
Hemarthrosis	Sign	HP:0005261 / HP:0003268	10/12 (83%)	Childhood
Umbilical stump bleeding	Sign	HP:0031364	Characteristic of severe neonatal disease	Neonatal
Intracranial hemorrhage	Sign	HP:0002170	Recurrent in severe disease; high morbidity	Neonatal–infancy
Subgaleal / subdural hematoma	Sign	HP:0002320	Reported in neonatal severe disease	Neonatal
Menorrhagia	Symptom	HP:0000132	Common in affected women	Adolescence–adult
Prolonged PT	Lab abnormality	HP:0008151	Universal	Congenital
Prolonged APTT	Lab abnormality	HP:0003645	Universal	Congenital
Reduced factor X activity	Lab abnormality	HP:0040189	Universal (diagnostic)	Congenital

Age of onset. Severe disease presents **neonatally**: *"Early neonatal bleeding, including umbilical and subgaleal hemorrhage, may be the initial manifestations of severe congenital FX deficiency, even in the absence of family history"* (PMID: 42144914). Milder deficiency may present in childhood or be detected incidentally in adulthood.

Severity and frequency. In a 12-patient severe cohort, *"the most frequent bleeding episodes in patients were epistaxis and easy bruising (11/12, 91%), followed by haemarthroses (10/12, 83%)"* (PMID: 26222694). Severity is graded by FX:C (see §8/§10).

Symptom progression. The bleeding tendency is **lifelong and episodic** — punctuated by spontaneous bleeds and provoked by trauma, surgery, or childbirth — rather than steadily progressive. Severity is generally stable for a given genotype.

Quality of life impact. Recurrent bleeds, joint damage from hemarthroses, menorrhagia, and the burden of prophylactic infusions affect daily functioning; disease-specific QoL data are limited, but the plain-language pdFX summary notes that patients' *"health and daily lives were impacted in different ways by HFXD"* (PMID: 42253376).

4. Genetic / Molecular Information

Causal gene. F10 (HGNC:3528; OMIM 613872), located on **chromosome 13q34-ter** (PMID: 30507709). It comprises 8 exons and encodes the ~488-residue mature factor X protein.

Protein domain architecture. FX is *"composed of the γ -carboxyglutamic acid (GLA) domain, two epidermal growth factor domains (EGF-1 and EGF-2), and the serine protease (SP) domain"* (PMID: 35059555); equivalently *"an N-terminal γ -carboxyglutamate (Gla) domain, two epidermal growth factor-like (EGF) domains, and a C-terminal trypsin-like serine protease (SP) domain"* (PMID: 30644641). The **vitamin K-dependent γ -carboxylation of GLA-domain glutamates** is required for calcium binding and phospholipid-membrane association.

Variant spectrum. An *"interactive FX variant database"* analyzed **180 genetic variants** across all four domains (GLA, EGF-1, EGF-2, SP), yielding genotype–severity insight (PMID: 35059555); HGMD previously cataloged **>149 F10 mutations**. Missense variants predominate. Representative pathogenic variants documented in the reviewed literature:

Variant (protein)	cDNA / exon	Domain	Effect	Reference
p.Val298Met	g.27881G>A, exon 8	SP	Homozygous; secondary-structure change; FX:C 1%	PMID: 27264807
p.Phe71Ser	c.212T>C, exon 2	GLA	Disrupts Ca ²⁺ -binding hydrogen bonds	PMID: 41451502
p.Val424Phe	c.1270G>T, exon 8	SP	Steric hindrance in catalytic domain	PMID: 41451502
p.Gln249Pro	homozygous	SP	FX:C <1%, severe bleeding	PMID: 42506896
p.Gly262Asp	—	SP	Recurrent; type II deficiency	PMID: 26222694
p.Leu487Phe	novel	SP	Iranian cohort	PMID: 35140190
p.Pro343Ser (FX Friuli)	—	SP	Dysfunctional; normal RVVT	PMID: 28030967

Variant classification. By ACMG/AMP criteria, most reported F10 variants are **pathogenic / likely pathogenic**; bioinformatic conservation analyses (e.g., Phe71, Val424 are highly conserved) support pathogenicity ([PMID: 41451502](#)).

Functional consequences. Two functional classes: - **Type I (quantitative)**: concordant reduction of FX activity and antigen (loss of protein). - **Type II (dysfunctional / qualitative)**: reduced activity with normal/near-normal antigen (defective protein). *"FX:Ag was reduced in all patients, consistent with type II deficiency"* in one cohort ([PMID: 26222694](#)); type II variants like FX Friuli show discordant assay behavior. The overarching molecular consequence is **loss of function** — reduced generation of active FXa.

Allele frequency. Individual pathogenic F10 alleles are very rare in gnomAD (consistent with disease prevalence of 1:500,000–1:1,000,000). Specific alleles cluster in consanguineous founder populations.

Somatic vs germline. All congenital variants are **germline**. (Somatic/acquired FX loss occurs in amyloidosis but is not genetic.)

Modifier genes / epigenetics / chromosomal abnormalities. No specific modifier genes, epigenetic mechanisms, or large-scale chromosomal abnormalities have been established as drivers of congenital FX deficiency; the disorder is essentially fully explained by F10 genotype plus vitamin K–dependent post-translational modification. Residual FX:C is the principal determinant of phenotype.

5. Environmental Information

Congenital FX deficiency is a **purely genetic disorder** with no environmental, toxic, lifestyle, or infectious cause. The relevant environmental factors are: - **Consanguinity** (sociocultural), which increases homozygosity and thus disease incidence in certain populations. - **Vitamin K availability and hepatic function**, which modulate FX synthesis and can aggravate the biochemical deficiency (FX is a vitamin K–dependent hepatic glycoprotein). - **No infectious agents** cause the congenital disease. (Historically, plasma-derived products carried viral-transmission risk, motivating high-purity/pathogen-reduced concentrates — a treatment-related, not disease-causing, consideration.)

6. Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic pathogenic F10 variant** (missense/frameshift/nonsense/splice/deletion) → **leads to** reduced synthesis (type I) or production of a dysfunctional FX protein (type II).
2. Reduced/defective FX → **results in** impaired vitamin K–dependent GLA-domain function and/or impaired serine-protease catalytic activity → **leads to** lower functional plasma FX and reduced conversion of zymogen FX to the active protease FXa.
3. Upstream, "*Tissue factor (TF) and factor VIIa (FVIIa) form the [extrinsic complex] together with FX on phosphatidylserine-containing membranes, leading to FX activation by TF:FVIIa*" (PMID: 39671302); the intrinsic FIXa/FVIIIa (tenase) complex activates FX in parallel. With defective FX, **both** activation routes yield less FXa. (*This is the branch point where extrinsic and intrinsic pathways converge on FX.*)

4. Reduced FXa → **results in** deficient assembly of the **prothrombinase complex** — *"the enzyme factor Xa (fXa), the cofactor fVa, Ca²⁺ and phospholipids, [which] activates the zymogen prothrombin to the protease thrombin"* (PMID: 35427420).
5. Deficient prothrombinase → **leads to** decreased **thrombin generation**.
6. Decreased thrombin → **results in** reduced fibrinogen-to-fibrin conversion, impaired platelet activation, and an unstable clot: *"FX plays a pivotal role in the coagulation cascade, activating thrombin to promote platelet plug formation and prevent excess blood loss"* (PMID: 35059555).
7. Impaired clot formation → **leads to** the clinical bleeding phenotype: umbilical, mucocutaneous, joint, and intracranial hemorrhage, with severity scaling inversely with residual FX:C.

Detail by category

- **Molecular pathways:** The **blood coagulation cascade** — extrinsic (TF:FVIIa), intrinsic (FIXa:FVIIIa tenase), and common pathway (prothrombinase). GO:0007596 (blood coagulation), GO:0007597 (intrinsic pathway), GO:0007598 (extrinsic pathway).
- **Protein dysfunction:** GLA-domain variants (e.g., p.Phe71Ser) disrupt Ca²⁺ binding and membrane association; SP-domain variants (e.g., p.Val424Phe, p.Val298Met) alter catalytic/structural integrity. Loss of function is the unifying consequence.
- **Biochemical abnormalities:** Deficiency of a serine protease (FXa; GO:0004252, serine-type endopeptidase activity) at the pivotal convergence of coagulation; requires vitamin K–dependent γ-glutamyl carboxylation (GO:0017187).
- **Cellular processes / cell types:** Coordinated on phosphatidylserine-exposing **platelet** (CL:0000233) surfaces and TF-bearing cells; **hepatocytes** (CL:0000182) synthesize FX. Structural work confirms FX engages membranes via its GLA domain (cryo-EM of prothrombin–prothrombinase, PMID: 35427420; membrane-bound TF:FVIIa:FX model, PMID: 39671302).
- **Tissue damage mechanism:** Not a primary tissue-destructive disease; damage is **secondary to hemorrhage** (intracranial bleeding → neurological injury; hemarthrosis → joint damage).
- **Immune involvement:** None in the congenital form. (Immune/autoimmune mechanisms characterize *acquired* FX deficiency — separate entities.)
- **Metabolic changes:** No systemic metabolic derangement; the "metabolic" defect is confined to vitamin K–dependent carboxylation of FX.

Upstream vs downstream: F10 genotype (upstream) → reduced FXa → reduced prothrombinase/thrombin (midstream) → impaired fibrin/platelet plug → bleeding (downstream clinical manifestation).

7. Anatomical Structures Affected

Body system: Hematologic / coagulation (blood) system — a systemic plasma-protein deficiency, so bleeding can affect any site.

Level	Structure	Ontology term
Organ (synthesis)	Liver (site of FX production)	UBERON:0002107
Fluid / tissue	Blood / plasma	UBERON:0000178
Primary clinical targets	Umbilical stump (neonate)	UBERON:0002331
	Brain / intracranial space	UBERON:0000955
	Scalp / subgaleal space	UBERON:0000403
	Joints / synovial cavity	UBERON:0002217
	Nasal mucosa (epistaxis)	UBERON:0001707
	Uterus / endometrium (menorrhagia)	UBERON:0000995
	Skin / subcutaneous tissue	UBERON:0002097
	Hepatocyte (synthesis)	CL:0000182
Cell types	Platelet (coagulation surface)	CL:0000233
	Extracellular region / plasma; PS-containing membranes	GO:0005576
	Hepatocyte ER (γ -carboxylation, secretion)	GO:0005783

Lateralization: Bleeding is site-dependent and typically not lateralized; intracranial bleeds may be focal.

8. Temporal Development

Onset. Congenital — the deficiency is present from birth. Severe disease presents in the **neonatal period** (umbilical, subgaleal, intracranial hemorrhage); moderate/mild disease may present in childhood or later. The onset pattern of individual bleeds is **acute/episodic** on a chronic constitutional background.

Severity classification by FX:C:

System	Severe	Moderate	Mild
Traditional	FX:C <1%	FX:C 1–5%	FX:C 6–10%
EN-RBD (revised)	FX:C <10%	FX:C 10–40%	FX:C >40%

"HFXD is traditionally classified by severity as severe (FX:C <1%), moderate (FX:C = 1%-5%), or mild (FX:C = 6%-10%)" (PMID: 41104456); the revised EN-RBD criteria define "severe (FX:C <10%), moderate (FX:C = 10%-40%), and mild (FX:C >40%)." Phenotype heterogeneity exists: *"Seven patients with FX:C >40% had bleeding episodes that required treatment, and 3 of them experienced multiple bleeding episodes" (PMID: 41104456).*

Progression / course. **Chronic, lifelong,** and **episodic/fluctuating** rather than progressive. Severity is stable for a given genotype. No spontaneous remission occurs; "remission" of the bleeding tendency is treatment-induced (factor replacement/prophylaxis).

Critical periods. The **neonatal/early-infancy window** is the period of greatest vulnerability to catastrophic intracranial hemorrhage and the key window for prophylactic intervention. Surgery, trauma, childbirth, and menstruation are recurring high-risk periods across life.

9. Inheritance and Population

Epidemiology. Ultra-rare. Estimated prevalence of severe disease is approximately **1 in 500,000 to 1 in 1,000,000** (HFXD affects 1:500,000–1:1,000,000 people worldwide; PMID: 29707881). In rare-bleeding-disorder cohorts, FX deficiency is a small minority of cases (e.g., 0.36% in a north-eastern Iran survey, PMID: 23114518; 4.2% of inherited coagulation defects in an Egyptian pediatric series, PMID: 22610136).

Inheritance. Autosomal recessive (PMID: 30507709). Affected individuals are homozygous or compound heterozygous; heterozygous carriers have ~50% FX levels and are usually asymptomatic.

Penetrance / expressivity. Biallelic pathogenic genotypes are essentially fully penetrant for a laboratory phenotype (reduced FX:C), but **clinical expressivity is variable** and correlates with residual FX:C. Heterozygotes with FX:C 40–50% show no bleeding tendency.

Genetic anticipation: Not applicable (not a repeat-expansion disorder).

Consanguinity and founder effects. Consanguinity is central: the disorder is enriched in consanguineous populations, and homozygous variants recur through consanguineous marriage (e.g., PMID: 27264807). Region-specific/founder variants occur (e.g., FX Friuli in an Italian population, PMID: 28030967; recurrent p.Gly262Asp, PMID: 26222694).

Population demographics. Higher prevalence in Iran, Turkey, Pakistan, Egypt, and other regions with consanguineous marriage. As an autosomal recessive trait, the **sex ratio is approximately 1:1**, though females carry additional bleeding burden from menorrhagia and obstetric hemorrhage.

10. Diagnostics

Screening coagulation tests. The diagnostic hallmark is **simultaneous prolongation of PT and APTT**, reflecting FX's position at the convergence of extrinsic and intrinsic pathways. *"Diagnosis was made by abnormal results of Coagulation factors screening mainly Prothrombin time, Activated partial thromboplastin time, Russell's viper venom test, mixing tests factor X assay"* (PMID: 9145616).

Confirmatory / classifying assays: - **FX coagulant activity (FX:C)** — one-stage clotting assay; the diagnostic and severity-grading measure. - **FX antigen (FX:Ag)** — ELISA; distinguishes **type I** (activity and antigen both reduced) from **type II** (activity reduced, antigen normal/near-normal). *"FX:Ag was reduced in all patients, consistent with type II deficiency"* (PMID: 26222694). - **Russell's viper venom time (RVVT)** and **mixing studies** — RVVT directly activates FX; useful for characterizing variants. Some dysfunctional variants behave atypically: FX Friuli shows *"prolonged partial thromboplastin time, prolonged prothrombin time but normal Russell viper venom clotting time"* (PMID: 28030967), and a rare variant with normal APTT and RVVT despite prolonged PT has been described (PMID: 4014297). Assaying FX through all three activation routes reveals molecular heterogeneity (PMID: 3970856).

Genetic testing. Direct **F10 sequencing** (all 8 exons and flanking regions) confirms the diagnosis, identifies the causal variant(s), and enables carrier/cascade and prenatal testing. Single-gene testing is sufficient given the monogenic etiology; rare-bleeding-disorder panels are an alternative. WGS/WES are rarely needed but can be used when the phenotype is ambiguous. Chromosomal microarray/karyotyping/FISH are not indicated (no chromosomal abnormality involved).

Differential diagnosis: - **Hemorrhagic disease of the newborn (vitamin K deficiency)** — a key mimic; severe congenital FX deficiency "*may be misdiagnosed as hemorrhagic disease of the newborn*" (PMID: 15036435). - **Acquired FX deficiency** — AL amyloidosis (mildly-to-severely reduced FX, e.g., PMID: 42622220), autoimmune/Sjögren-associated (PMID: 42227468), and transient inhibitor-mediated (PMID: 9495379) forms; distinguished by later onset, absent family history, associated systemic disease, and (for inhibitors) incomplete correction on mixing. - Other common-pathway factor deficiencies (FV, FII), combined VKD-factor deficiency, and vitamin K antagonist effect.

Screening for asymptomatic individuals. Cascade/carrier testing within affected families (via the known familial F10 variant) and prenatal diagnosis are the applicable approaches; there is no routine population newborn screen for FX deficiency.

11. Outcome / Prognosis

Mortality / morbidity. The chief threat to life and long-term function is **intracranial hemorrhage**, particularly in severe neonatal disease. Untreated severe disease historically carries a poor prognosis; an infant with severe FX deficiency had "*three intracranial hemorrhages in the first 6 months of life*" before prophylaxis (PMID: 1530121), and siblings with severe disease presented with "*severe intracranial bleeding*" in which plasma replacement was not efficacious (PMID: 15036435).

Effect of treatment. Prognosis is transformed by prophylaxis: prophylactic prothrombin complex infusions "*may prevent the poor outcome usually described in these patients*" (PMID: 1530121), and modern high-purity pdFX prophylaxis provides excellent bleed prevention (see §12). With adequate replacement, life expectancy approaches normal and joint/neurological morbidity is markedly reduced.

Disease course / complications. Lifelong bleeding risk; complications include **hemophilic arthropathy** from recurrent hemarthrosis, **neurological sequelae** from ICH, obstetric hemorrhage and miscarriage in affected women, and, in resource-limited settings, transfusion-transmitted infection from older plasma products.

Prognostic factors. Residual FX:C is the dominant prognostic biomarker — *"Mutations leading to a FX:C of less than 1% are associated with severe bleeding symptoms confirming the strong correlation between clinical severity and FX:C"* (PMID: 35140190); *"Severe bleeding in FX deficiency is associated with FX:C <5%, while milder deficiencies often present minimal symptoms"* (PMID: 42506896). Genotype (homozygous/compound-heterozygous; GLA/SP-domain variants) and early ICH history further stratify risk. Access to specific factor replacement is a major determinant of outcome.

12. Treatment

First-line: high-purity plasma-derived FX concentrate (pdFX / Coagadex). pdFX is *"the first single-factor replacement therapy indicated for hereditary FX deficiency"* and is *"a safe and efficacious treatment option in patients aged ≥ 12 years with hereditary FX deficiency"* (PMID: 27797267), approved in the US and EU for **on-demand treatment, prophylaxis, and perioperative management**. Phase 3 trials rated efficacy excellent with no inhibitor development, including in children <12 years (PMID: 29707881), women/girls (PMID: 29460388), and specific cohorts (PMID: 29545231).

Prophylaxis dosing. *"Routine prophylaxis with pdFX be initiated at 25 IU/kg twice weekly in adults/adolescents ≥ 12 years of age, and at a dosage of 40 IU/kg twice weekly in children <12 years of age"* (PMID: 35499465). On-demand dosing targets FX:C >5 IU/dL for hemostasis.

Alternatives / adjuncts (tiered). *"In centers where FX concentrate is unavailable, PCC and FFP provide effective hemostatic coverage"* (PMID: 41836993) — demonstrated in perioperative pseudotumor evacuation. Real-world management combines modalities: *"treatments included antifibrinolytics, hormonal therapy, and factor replacement"* (PMID: 41104456).

Tier	Agent	NCIT-type term	Role / notes
1st line	Plasma-derived FX concentrate (pdFX/ Coagadex)	Coagulation Factor X (Human)	Specific single-factor replacement; on-demand, prophylaxis, perioperative
Alternative	Prothrombin complex concentrate (PCC)	Prothrombin Complex Concentrate	Contains FX; thrombotic risk; use when pdFX unavailable
Alternative	Fresh frozen plasma (FFP)	Fresh Frozen Plasma	Broad replacement; volume-overload risk
Adjunct	Antifibrinolytics (tranexamic acid)	Tranexamic Acid	Mucosal bleeds, menorrhagia, dental/surgical
Adjunct	Hormonal therapy	Hormone Therapy	Menorrhagia management in women

Perioperative and obstetric management. Factor replacement targets hemostatic FX:C perioperatively; obstetric care requires prophylaxis given elevated miscarriage/antenatal-hemorrhage rates (§13).

Treatment outcomes / safety. pdFX has minimal side effects, high treatment-success rates (98% in women/girls; PMID: 29460388), and no inhibitor development reported. PCC carries thrombotic risk and FFP carries volume-overload and (for non-pathogen-reduced products) infection risk.

Pharmacogenomics / advanced therapeutics. No pharmacogenomic dosing guidance is established. **Gene therapy** is not yet clinically available for FX deficiency but is conceptually attractive (single, hepatically expressed gene); zebrafish and mouse models support target validation (§14–15).

13. Prevention

Because congenital FX deficiency is genetic, prevention is **reproductive and genetic**, not lifestyle-based.

- **Genetic counseling and carrier/cascade testing:** As an autosomal recessive disorder enriched in consanguineous families, identifying carriers via the familial F10 variant is central.
 - **Prenatal and preimplantation diagnosis:** *"Advances in preconception genetic counselling and prenatal diagnosis, including non-invasive prenatal testing (NIPT) and preimplantation genetic diagnosis (PGD), are crucial for informed reproductive choices and delivery planning"* (PMID: 41988968).
 - **Reproductive/obstetric management (secondary prevention):** FX deficiency and *"other severe factor deficiencies ... are associated with higher rates of miscarriage and antenatal haemorrhage, often requiring prophylactic factor replacement"* (PMID: 41988968). Mode-of-delivery decisions aim to minimize cranial bleeding risk in an affected fetus.
 - **Tertiary prevention (preventing complications):** Prophylactic factor replacement prevents recurrent intracranial hemorrhage and hemarthropathy (PMID: 1530121).
 - **Immunization / public-health / behavioral prevention:** Not applicable to a monogenic coagulopathy (beyond general trauma-avoidance counseling and hepatitis B vaccination for patients receiving blood products).
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14. Other Species / Natural Disease

Naturally occurring disease. Congenital FX deficiency occurs spontaneously in non-human species, supporting cross-species conservation of the coagulation pathway. In a domestic cat: *"Severe congenital deficiency of factor X was diagnosed in a 3-year-old castrated male domestic shorthair cat with clinical signs of generalized seizures and prolonged bleeding after venipuncture"* (PMID: 9290823); the seizures reflected intracranial bleeding, and reduced FX activity plus prolonged RVVT in the dam and a sibling indicated heritability — a natural phenocopy of severe human disease.

Taxonomy / orthologs. F10 is conserved across vertebrates; orthologs include mouse *F10* (NCBI Gene ID 14058), rat *F10*, zebrafish *f10*, and cat *F10*. The vitamin K–dependent coagulation factor family is evolutionarily ancient (PMID: 30644641).

Comparative biology. Mammals (cat, mouse) show severe/lethal phenotypes with FX deficiency, whereas zebrafish tolerate severe common-pathway defects better: *"Deficiency of factor X (F10) in humans is a rare bleeding disorder with a heterogeneous phenotype and limited therapeutic options"* was studied via genome editing in zebrafish, which revealed *"unexpected tolerance of severe defects in the common pathway"* (PMID: 28576875). This species difference is informative about pathway redundancy.

Zoonotic potential: None (genetic disorder).

15. Model Organisms

Mouse (mammalian) knockout — closest phenotypic model. Targeted deletion of the exons encoding mature FX produces a faithful model of severe human disease: *"homozygous deficiency results in partial embryonic lethality at embryonic day (E) 11.5-12.5 with signs of massive bleeding"* and *"the majority of those that survive to term die within 5 days, most frequently from intraabdominal bleeding"* (PMID: 12161341); survivors die between P5–P20 from intraabdominal, subcutaneous, or intracranial bleeding. Comparative knockout work situates FX among coagulation factors in development: *"Factor X (FX) deficiency causes partial embryonic lethality between E11.5-12.5. FX^{-/-} mice that were born died from fatal neonatal bleeding"* (PMID: 11841337).

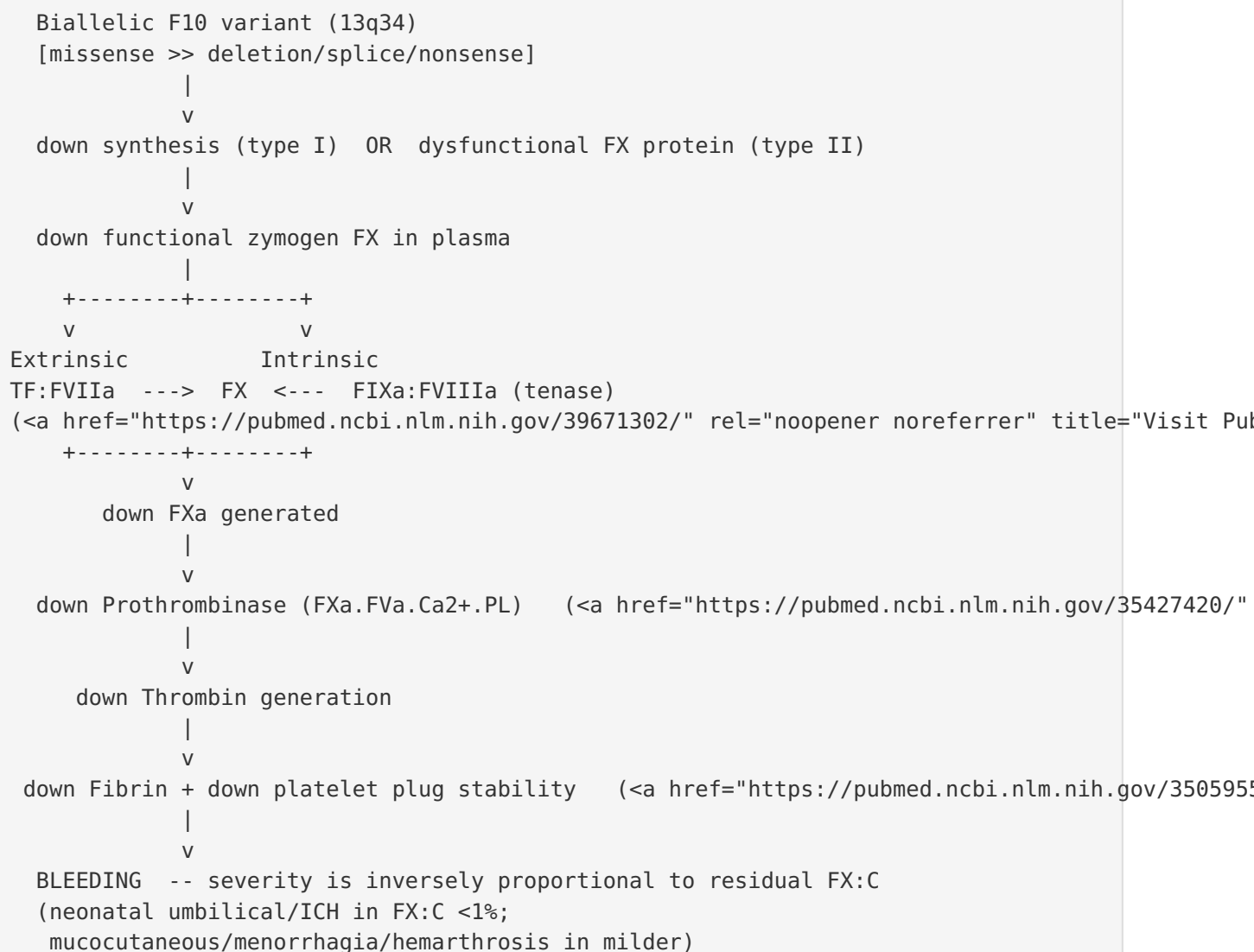
Model recapitulation and limitations. The mouse knockout **recapitulates fatal neonatal/perinatal hemorrhage** (including intracranial bleeding), mirroring severe human disease. Its principal limitation is **embryonic/perinatal lethality of the complete null**, precluding study of chronic adult disease; hypomorphic/knock-in alleles (e.g., FX Friuli chimeric mice) are needed to model milder, survivable phenotypes.

Zebrafish. CRISPR/genome-edited *f10* zebrafish are available and reveal that *"severe defects in the common pathway"* are unexpectedly tolerated (PMID: 28576875) — useful for therapeutic screening and pathway-redundancy studies, but a weaker phenocopy of the mammalian bleeding phenotype.

Applications. These models support study of coagulation-pathway biology, FX's developmental roles beyond hemostasis, and preclinical testing of replacement and gene-therapy strategies.

Model resources: MGI (mouse *F10*), ZFIN (zebrafish *f10*), Alliance of Genome Resources.

Mechanistic Model / Interpretation



The central organizing principle is a **dose–response relationship between residual FX coagulant activity and bleeding severity**. Because FX is the single non-redundant convergence node of both coagulation pathways, even partial loss produces a measurable dual PT/APTT prolongation, and profound loss (<1%) produces catastrophic neonatal hemorrhage. Every downstream clinical, diagnostic, prognostic, and therapeutic feature flows from this quantitative relationship: FX:C is simultaneously the diagnostic analyte, the severity classifier, the prognostic biomarker, and the treatment target (maintain FX:C >5 IU/dL).

Evidence Base

PMID	Contribution	Evidence type
30507709	AR inheritance, F10/13q34, variant spectrum (22 mutations)	Human, multicentre cohort
35059555	Domain architecture; 180-variant database; FX's role in thrombin/plug	Human, database/review
35140190	FX:C <1% ↔ severe bleeding; novel Leu487Phe	Human cohort (Iran)
42506896	FX:C <5% severe threshold; p.Gln249Pro	Human case series
42144914	Neonatal umbilical/subgaleal presentation	Human case report
26222694	Phenotype frequencies; type II via FX:Ag; p.Gly262Asp	Human cohort
41104456	Traditional & EN-RBD severity classes; multimodal treatment	Human case series
12161341	FX−/− mouse embryonic lethality + neonatal bleeding	Model organism
11841337	Comparative coagulation-factor knockouts	Model organism
27797267	pdFX as safe/effective single-factor therapy	Human trial
35499465	pdFX prophylaxis dosing	Human review
29707881	pdFX in children <12; prevalence 1:500k–1M	Human trial
9145616	Diagnostic assay panel (PT/APTT/RVVT/FX assay)	Human case
28030967	FX Friuli type II variant; atypical RVVT	Human
35427420	Cryo-EM prothrombin–prothrombinase (mechanism)	Structural
39671302	Membrane TF:FVIIa:FX extrinsic complex (mechanism)	Structural/computational
9290823	Natural feline FX deficiency	Veterinary
28576875	Zebrafish f10 model; pathway tolerance	Model organism
1530121	Prophylaxis prevents recurrent ICH	Human case
15036435	ICH in siblings; HDN misdiagnosis	Human case
41988968	Reproductive risk, NIPT/PGD prevention	Human review

PMID	Contribution	Evidence type
41836993	PCC/FFP as alternatives	Human case
30644641	VKD domain architecture; family evolution	Review

Limitations and Knowledge Gaps

- 1. Rarity limits statistics.** All human evidence derives from case reports and small cohorts (n = 6–24); no large randomized trials or population-scale registries with hard survival endpoints exist. Frequency figures (e.g., 91% epistaxis) come from single small cohorts and may not generalize.
- 2. Genotype–phenotype resolution is incomplete.** Although FX:C predicts severity well, the phenotypic heterogeneity among FX:C >40% patients ([PMID: 41104456](#)) indicates unexplained modifiers; variant-level functional data are sparse for many alleles.
- 3. No modifier genes / epigenetics established** — a genuine gap rather than a negative finding.
- 4. Model limitations.** The complete FX-null mouse is perinatally lethal, precluding adult-disease modeling; zebrafish tolerate the defect, weakening phenocopy fidelity. Hypomorphic mammalian models for milder disease are underdeveloped.
- 5. No dedicated omics.** No transcriptomic, proteomic, or metabolomic signatures specific to congenital FX deficiency were found; the disorder is defined at the single-gene/single-protein level.
- 6. Prognostic quantification.** Precise mortality/ICH incidence rates and long-term QoL metrics under modern prophylaxis are not well quantified in the reviewed literature.
- 7. Gene therapy absent.** No clinical gene-therapy data for FX deficiency yet exist despite biological plausibility.

Proposed Follow-up Experiments / Actions

1. **Curate a comprehensive genotype–phenotype matrix** from the interactive FX variant database ([PMID: 35059555](#)), mapping domain-specific variants (GLA vs EGF vs SP) to FX:C and bleeding scores, to explain the FX:C >40% heterogeneity.
 2. **Develop hypomorphic / knock-in mouse models** (allelic series reproducing FX:C of ~1%, 5%, 10%) to study survivable chronic disease and test prophylaxis regimens.
 3. **Preclinical AAV gene-therapy studies** targeting hepatic F10 expression, using established mouse/zebrafish models to test durability and hemostatic correction.
 4. **Prospective natural-history registry** with standardized bleeding scores (ISTH-BAT), FX:C, genotype, and QoL instruments (SF-36/PROMIS) to quantify prognosis under modern pdFX prophylaxis.
 5. **Functional characterization of type II variants** (activity-normal-antigen) via recombinant expression and thrombin-generation assays to refine mechanistic classification.
 6. **Population carrier-frequency studies** in consanguineous populations using gnomAD-anchored F10 allele frequencies to inform screening programs.
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Consensus Answer

Congenital Factor X deficiency (MONDO:0009212; OMIM 227600) is an ultra-rare (~1:500,000–1:1,000,000) autosomal recessive bleeding disorder caused by biallelic, predominantly missense pathogenic variants in **F10** (chromosome 13q34) encoding the vitamin K–dependent serine protease factor X, which sits at the convergence of the extrinsic and intrinsic coagulation pathways. Reduced or dysfunctional FX lowers FXa/prothrombinase assembly and thrombin generation, producing a bleeding phenotype whose severity correlates inversely with residual FX coagulant activity (FX:C): severe disease (FX:C <1%) presents neonatally with umbilical-stump and intracranial hemorrhage, while milder forms cause mucocutaneous bleeding, menorrhagia, and hemarthrosis. It is diagnosed by concurrently prolonged PT and APTT with a confirmatory reduced FX:C assay (FX:Ag distinguishing type I from type II) and managed with factor replacement—high-purity plasma-derived FX concentrate (pdFX/Coagadex) first-line, or PCC/FFP plus antifibrinolytic and hormonal adjuncts—where prophylaxis prevents the otherwise poor outcome of recurrent intracranial hemorrhage.

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